

The logic of the English auxiliary system

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Abstract

This paper proposes an analysis of English auxiliary verbs in Hybrid Type-Logical Grammar. The proposed analysis integrates the key analytic ideas of both transformational and nontransformational approaches in the literature within a logic-based architecture of grammar in which the notion of (formal) inference plays a central role. We show that the proposed analysis not only provides principled and explicit analyses of major properties of English auxiliary verbs but also has interesting implications for crosslinguistic and diachronic aspects of the auxiliary system.

Keywords: auxiliary verb, *do* support, NICE properties, Type-Logical Grammar, higher-order auxiliary

1 Introduction

The syntax of auxiliary verbs in English has served as one of the most important touchstones throughout the history of generative syntax. It was a key component in Chomsky’s (1957) argument for the notion of syntactic transformations, played a pivotal role in the argument for the viability of a nontransformational, lexicalist version of generative syntax in GPSG research (Gazdar et al. 1982), and was recently invoked in an argument for a ‘constructional’ view of grammar embodied in the current version of HPSG (Sag et al. 2020). Moreover, having a precise understanding of the auxiliary system is important in broader linguistic research too, since some aspects of it (most notably the subject-auxiliary inversion pattern) has been the target of a long-standing controversy on the innateness of linguistic knowledge (see, e.g., Pullum and Scholz (2002); Berwick et al. (2011); Estigarribia (2010), and a recent summary by Chaves and Putnam (2020)), and related attempts at computational modelling of language learning/induction (Borensztajn et al. 2009; Warstadt and Bowman 2020). Yet despite the abundance of research on this topic in the syntactic literature, the complexity of data in this domain still defies a completely successful analysis, with two key issues, one morpho-syntactic and the other semantic, posing significant challenges to any approach: the so-called ‘*do* support’ facts and the interactions between modal auxiliaries and other scopal operators.

The present paper takes up this challenge, and offers a new analytic angle from the perspective of a logic-based architecture of grammar embodied in Hybrid Type-Logical Grammar (Hybrid TLG; Kubota and Levine 2015, 2020). We argue that a conceptually

simple and formally explicit analysis of both the ‘*do* support’ pattern and the scopal properties of auxiliary verbs immediately emerge from this logic-based perspective. The analysis incorporates (what we take to be) the key insights of both transformational and nontransformational analyses of auxiliaries, without inheriting the shortcomings of either. Finally, interesting crosslinguistic and diachronic implications follow from our approach. These results, we argue, strongly suggest that taking logic as a fundamental principle underlying natural language syntax has much to offer to contemporary linguistic research.

The paper is structured as follows. Section 2 reviews the familiar facts about English auxiliaries, followed by a brief sketch of past literature in both the transformational and nontransformational approaches. Section 3 then presents our own analysis, by emphasizing their connections to familiar ideas in both transformational and nontransformational syntax. Section 4 is devoted to crosslinguistic and diachronic implications, and section 5 concludes the paper.

2 English auxiliary verbs: facts and literature

2.1 Key empirical properties of English auxiliary verbs

Auxiliaries are commonly characterized as members of a natural class whose distributional characteristics are captured by their occurrence in the so-called ‘NICE’ environments (Negation, Inversion, Contraction and Ellipsis).

- (1) a. John {will/should} buy the book.

- b. John {will/should} not buy the book. (cf. *John buys not the book.)
- c. {Will/Should} John buy the book? (cf. *Buys John the book?)
- d. John {won't/shouldn't} buy the book. (cf. *John buysn't the book.)
- e. Who will buy the book? – John {will/should}. (cf. *John buys.)

The distribution of the unstressed form of *do* is especially important in this connection, as it has played a non-negligible role in the history of generative grammar. As is well-known, unstressed *do* (borrowing Sag et al.'s (2020) notation, we write this as *dŏ* in what follows) appears in all the NICE environments but not in simple declarative sentences:

- (2)
- a. *John {dĭd/dŏes} buy the book.
 - b. John {dĭd/dŏes} not buy the book.
 - c. {Dĭd/Dŏes} John buy the book?
 - d. John {dĭdn't/dŏesn't} buy the book.
 - e. Who {bought/buys} the book? – John {dĭd/dŏes}.

These patterns were taken to provide strong empirical support for the so-called *Do* Insertion transformation in early transformational syntax (Chomsky 1957). By contrast, the *do* support paradigm has long remained problematic in nontransformational treatments of English auxiliaries (see, e.g., Sag et al. 2020) since such approaches treat *do* essentially on a par with other auxiliaries.

The ‘*do* support’ facts pertain primarily to the morpho-syntactic aspect of auxiliary verbs. Another aspect that poses significant analytic challenges to many extant approaches

is their semantic properties, especially, scopal interactions with other operators. A relatively simple version of such scopal interaction can be observed with quantifiers in the subject position:

- (3) (The Board's decision means that) every student can vote.

This sentence has a reading in which the modal auxiliary *can* outscopes the subject position quantifier *every student*—on this reading, the sentence is making a claim about studentship and the right to vote, not about the (voting) rights of individuals who happen to be students in the actual world. Such a scopal relation is straightforward in an analysis in which the modal occupies a position that outscopes the quantifier at the level of representation relevant for semantic interpretation, but necessitates some complication in a surface-based analyses typically entertained in the G/HPSG and categorial grammar literature.

A similar pattern of scopal interaction can be observed with data such as the following in which modals interact with Gapping, originally noted by Dick Oehrle and Muffy Siegel (Oehrle 1971; Siegel 1984, 1987; Oehrle 1987).

- (4) a. John shouldn't eat steak and Mary $\emptyset$ just (eat) pizza.
b. Kim wouldn't play bingo or Sandy $\emptyset$ sit at home all evening.

(4a) has a reading in which what is prohibited is for two things to happen at the same time: John eating steak and Mary eating pizza. Note in particular that if we simply 'restore' the omitted (auxiliary) verb *shouldn't (eat)* in the second conjunct, the sentence means something totally different. In that non-gapping variant (...and Mary shouldn't...), the

sentence says that neither of the two things should happen: John eating steak and Mary eating pizza. Similarly, (4b) has a reading in which neither of the two things will happen ($\neg(K \vee S) \equiv \neg K \wedge \neg S$). This again is surprising on the view that Gapping simply involves omission of some expression recoverable from the preceding clause.

2.2 Pollock's (1989) Head Movement account

In the rest of this section, we summarize major treatments of English auxiliaries in transformational syntax (this subsection) and nontransformational syntax (next subsection). As is well-known, the English auxiliary system including the NICE properties was one of the key components of Chomsky's (1957) argument for transformational grammar. However, the specific analysis sketched by Chomsky in that work involved several ad-hoc features inherent to the early form of transformational grammar. Subsequent work thus attempted to eliminate the stipulative aspects of Chomsky's (1957) proposal. These efforts culminated in the influential proposal by Pollock (1989), in which all the NICE properties were reinterpreted as consequences of movement of certain items to Infl or still higher positions in clausal structure.

The gist of Pollock's (1989) analysis can be illustrated by the following configurational skeleton:

- (5) [CP C [TP [T' T [NegP [Neg *not*] [AgrP Agr [VP (Adv) V]]]]]]

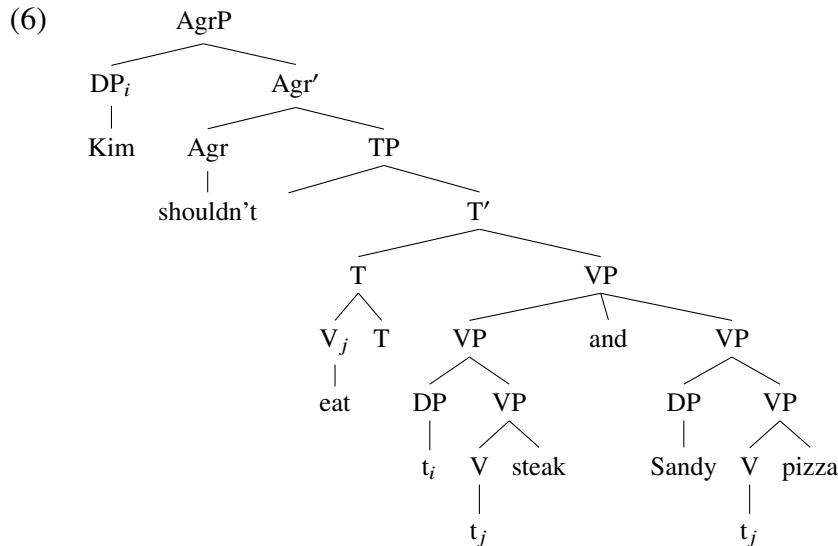
Pollock further makes a couple of key assumptions about the 'origins' of different lexical items:

- auxiliaries, which are functional elements, are generated at the head of Tense
- main verbs are generated under V
- among the main verbs, *have* and *be* are the only ones that can move to the head of Agr, due to the lack of a θ grid

It then follows that inversion is restricted to modals and *have* and *be* since movement to C (the landing site for the ‘inverted’ auxiliary) through T is licit only from the Agr position, and not from the lower lexical V position directly. For lexical verbs, Pollock assumes a lowering movement of the tense affix. The same kind of reasoning applies to the possibility of preceding *not*, i.e., appearing above NegP. Only those elements that can undergo Head Movement to Agr can move beyond NegP, which again restricts sentential negation to modals and aspectual auxiliaries. Thus, the NIC(E) properties fall out for items that share the property of being able to occupy the head of T, either via movement, or base generation. In short, Pollock’s strategy here is to derive the common behavior of the modals, *have*, and *be* via a complex network of abstract assumptions, most of which are predicated in terms of configurational notions referring to the structure in (5).

This stance is evident in the treatment of the auxiliary *do* as well, which, in contrast to both the aspectual verbs and the modals, originates under Agr as an optional possibility. Cooccurrence with *do* is ruled out in the case of *be* or *have* (**do have V-ed/do be V-ing*) due to some form of presumed semantic ill-formedness, and in the case of modals, due to the failure of a modal occupying the T position to bind a variable in the Agr position, since the latter is occupied by the overt form *do*.

An interesting application of Pollock's ideas can be found in Johnson's (2000) analysis of anomalous scope in Gapping sentences (cf. (4) above). The analysis is illustrated in the following tree diagram:¹



Johnson assumes the VP-internal subject hypothesis (Zagona 1982). Gapping is then analyzed via ATB movement of the verb and non-ATB movement of the subject of the first conjunct, both out of the VP-internal positions. Crucially, following Pollock (1989), modals originate under the root Agr, which means that they are *already* in a position to scope over the VP-level lower conjunction. While this represents an ingenious attempt to explain the anomalous scope in Gapping by independently motivated principles, the low-VP coordination structure that Johnson assumes has a number of empirical problems (see, e.g., Kubota and Levine (2016)).

Just as with any influential idea, Pollock's proposal has been challenged on multiple

¹Unlike Pollock, Johnson assumes that Agr is above T. While the difference may have important theoretical consequences within P&P analyses (see, e.g., footnote 19 in Pollock (1989, 384)) we do not delve into this issue here since this theory-internal issue has no ramifications for our own analysis.

fronts (see, e.g., Kim and Sag (2002)), and such challenges are often well-grounded: there are many aspects which are supposedly independently motivated but whose details remain unworked out, such as the often criticized ‘downward movement’ reformulation of Affix Hopping and the lack of θ grid for *be* and *have*. Nonetheless, we find it important not to miss its primary import in having in effect thrown down a gauntlet to competing approaches: is there any alternative framework in which there is some analogue to P&P’s unification of the NICE properties in terms of Head Movement, which motivates the exclusiveness of the NICE properties to the auxiliaries?

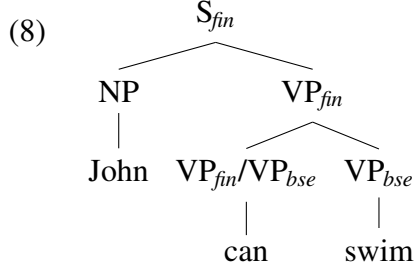
2.3 Monostratal accounts in lexicalist syntax

In lexicalist theories such as HPSG and categorial grammar, there is a long tradition of treating modal auxiliaries as expressions that take a VP as an argument to return another VP (Bach 1980; Gazdar et al. 1982). This analysis is adopted in major proposals in both HPSG (Pollard and Sag 1994) and Combinatory Categorical Grammar (Steedman 1996).

On this approach, which we illustrate below in a categorial grammar notation, the modal auxiliary *can* has the following type of lexical entry (we adopt a notation in which a tripartite linguistic sign consists of the prosodic representation, semantic interpretation and syntactic category):

$$(7) \quad \text{can}; \lambda P \lambda x. \Diamond P(x); \text{VP}_{fin} / \text{VP}_{bse}$$

and a sentence containing a modal *can* be licensed as follows:



The tree notation in (8) can be rewritten in the proof tree notation that is more standard in categorial grammar:

$$(9) \quad \frac{\frac{\text{can}; \lambda P \lambda x. \Diamond P(x); VP_{fin}/VP_{bse} \quad \text{swim}; \mathbf{swim}; VP_{bse}}{\text{can} \bullet \text{swim}; \lambda x. \Diamond \mathbf{swim}(x); VP_{fin}} /E \quad \text{John}; \mathbf{j}; NP}{\text{john} \bullet \text{can} \bullet \text{swim}; \Diamond \mathbf{swim}(\mathbf{j}); S_{fin}} \backslash E$$

As should be clear from this proof tree, (7) essentially says that *can* takes a base form VP as an argument to return a finite VP, where *fin* and *bse* are syntactic features that distinguish different morphological forms of verbs. This ensures the morpho-syntactic dependency between the higher and lower verbs (i.e., the ‘Affix Hopping’ effect; see Blevins and Sag (2013) for a lucid exposition). The semantics for the modal auxiliary says that it takes a property P and an individual x as arguments and applies the propositional operator $\Diamond$ to the proposition formed by giving x as an argument to P . Thus, the right meaning can be obtained without having the modal syntactically scope over the sentence in which it occurs.

In contrast to the structure-driven analysis in transformational approaches (such as Pollock (1989)), lexicalist approaches account for the NICE properties by means of lexical operations (or their analogs). For example, VP ellipsis can simply be analyzed by a valence reduction rule like the following:

$$(10) \quad \varphi; \mathcal{F}; VP_f/VP_b \Rightarrow \varphi; \mathcal{F}(P); VP_f$$

where P designates some property salient in the discourse. This is a version of anaphoric (or ‘interpretive’) analysis of ellipsis that does not involve an actual deletion of syntactic object. While we omit the details, inversion ($VP_f/VP_b \Rightarrow S_f/NP/VP_b$), negation and contraction can be given similar lexical treatments.

Simple and attractive as it may appear, this lexicalist approach faces significant challenges in view of more complex types of data of the sort noted in section 2.1. For example, the following sentence has a reading in which the modal auxiliary outscopes the subject position quantifier.

(11) (The Board’s decision means that) every student can vote. ($\Diamond > \forall/\forall > \Diamond$)

The lexical entry for the modal in (12) yields only the quantifier wide scope reading, and does not license the modal wide scope reading. One commonly adopted solution, originally due to Bach (1980), is to posit the following type of additional lexical entry for the modal which subcategorizes for a GQ-type expression (of type $(e \rightarrow t) \rightarrow t$) in the subject position:

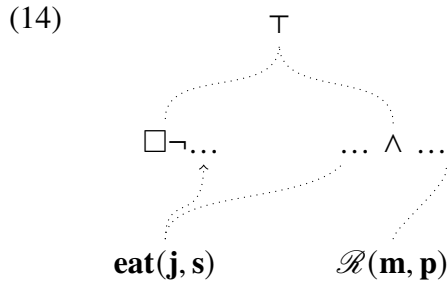
(12) can; $\lambda f \lambda \mathcal{F} . \Diamond \mathcal{F}(f); ((S_f/VP_f) \setminus S_f)/VP_b$

This essentially amounts to a claim that the scope ambiguity in (11) is a type of lexical ambiguity rather than a syntactic ambiguity (see also Pollard and Yoo (1998) for a different approach utilizing Cooper’s (1983) ‘storage’ mechanism).

While the ambiguity with subject position quantifiers may be amenable to a lexical treatment of the sort illustrated above, the scope interactions with Gapping (also discussed in section 2.1) turn out to be much trickier for strictly lexicalist approaches:

- (13) a. John shouldn't eat steak and Mary $\oslash$ just (eat) pizza.
 b. Kim wouldn't play bingo or Sandy $\oslash$ sit at home all evening.

Lexicalist approaches have remained silent on cases such as (13) until quite recently, but Park et al. (2019) (see also Park (2019)) explicitly addresses this issue. In their analysis, the lexical entries of the clause-level conjunction words *and* and *or* are underspecified as to the relative scope between the propositional operator contributed by the modal auxiliary in the first conjunct and the boolean conjunction/disjunction operator contributed by the conjunction word itself. The gist of the proposal can be schematically shown in (14):



The underspecification represented by the dotted lines can be resolved in the following two ways:

- (15) a. $\square \neg [\text{eat}(\mathbf{j}, \mathbf{s}) \wedge \text{eat}(\mathbf{m}, \mathbf{p})]$
 b. $\square \neg \text{eat}(\mathbf{j}, \mathbf{s}) \wedge \square \neg \text{eat}(\mathbf{m}, \mathbf{p})$

The key idea here is the use of underspecification to relax the mapping between surface syntax and final semantic interpretation. While this is a rather skillful use of a familiar technique in computational linguistics to a recalcitrant linguistic problem, what is left unaddressed is the question of why Gapping exhibits this type of scope ambiguity and what

motivates the underspecified semantics for modal and conjunction operators that Park et al. (2019) assume.

Finally, the *do* support facts remain a challenge for lexicalist approaches. In such approaches, *do* is just another auxiliary verb having exactly the same syntactic property as modals. But then, there is no obvious way to restrict its distribution to just the NICE environments. Sag et al. (2020) offers an explicit analysis of *do* support facts in the broader non-transformational literature, which departs from the strongly lexical approach in the G/HPSG tradition and relegates some of the explanatory burden to the constructional component. Their analysis essentially stipulates that *do* is lexically specified as a special type of auxiliary that can appear in only the NICE constructions, via a diacritic syntactic feature AUX (which no longer retains its original transparent function of identifying auxiliary verbs as a lexical class).

To summarize, while there have been important proposals in the recent literature addressing the two types of empirical challenges to nontransformational approaches to English auxiliaries, the proposed solutions do not necessarily attempt to derive these properties from more basic principles of grammar in ways that transformational approaches have attempted to do.

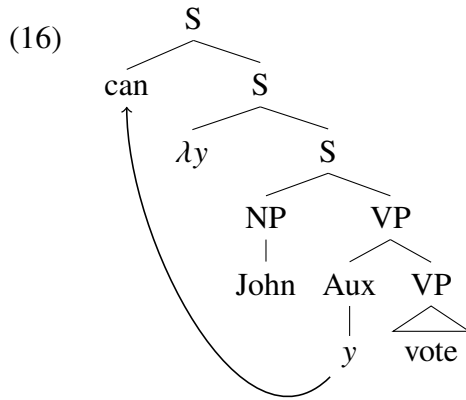
3 A type-logical analysis of English auxiliaries

The upshot of the discussion above is that both the transformational and the nontransformational traditions overemphasize one of the two criteria for successful theoretical research at

the expense of the other: attention to detail and an aspiration for a general and principled account. The goal of the rest of this paper is simple. We will try to convince the reader that this is indeed a false dichotomy. But in order to see this point, we need to take a slightly different perspective on the architecture of grammar from what we have been familiar with: instead of viewing syntax as a system of structure building, we propose to view it as a system of logical inference. This is in fact a view inherent to Type-Logical Grammar research in general (Lambek 1958; Moortgat 1997). And just as in other empirical domains (see Kubota and Levine 2016, 2017, 2020), it offers a unique synthesis of important analytic ideas on the English auxiliary system from both the transformational and nontransformational traditions, resulting in an analysis that does not compromise rigor and attention to detail by being fully general and principled.

3.1 Modals as scope-taking operators

Our analysis borrows from the Pollock/Johnson GB account the idea that modal verbs are semantic operators that take scope over a proposition. But we implement it in a slightly different manner. Specifically, we adopt an analysis of modal auxiliaries whose key idea can be schematically represented in the informal figure shown in (16), where the auxiliary verb occupies the preverbal position in the overt string—where it would ordinarily appear under the VP/VP string version—but is interpreted as a wide-scoping operator. This analysis does the work intended by the transformationalist ‘covert movement at LF’ analysis via altogether ordinary composition in our type-logic, with no special machinery or operations:



We wish to emphasize that the picture in (16) should not be taken in any way literally. As will become clear in the ensuing discussion, despite sharing some of the key analytic ideas at the broader level (as should be clear from the above exposition), there are a number of important differences at the foundational level between our proposal and the treatments of modal auxiliaries in the transformational literature (such as Iatridou and Zeijlstra (2013)) both conceptually and technically. Most importantly, our approach predicts the scopal interactions between modals and other elements as a logical consequence of the calculus, where the simpler VP/VP analysis of modal auxiliaries in lexicalist approaches falls out as a strict theorem. In a sense, our approach offers a way to unify the transformational treatment of auxiliaries as having scoping possibilities seemingly at odds with their overt position (e.g., Pollock 1989; Johnson 2000) with the monostratal analyses pursued in phrase-structure-theoretic approaches (e.g., Gazdar et al. 1982, 1985; Sag et al. 2020).

In what follows, we first illustrate the technical setup of Hybrid TLG with the more familiar type of scope-taking expressions, namely, NP-type quantifiers such as *every N* and *no N*.² We then extend this analysis to modal auxiliaries, with a slight change in lexical

²The complete rule system of Hybrid TLG is given in Appendix A. See Kubota and Levine (2020, chapter

assignment. The analysis of quantifiers utilizes the lambda calculus for characterizing the prosodic component of linguistic expressions, an idea due to Oehrle (1994). (17) is then analyzed as in (19), with the entry for the quantifier in (18).

(17) John read every book.

(18) $\lambda\sigma.\sigma(\text{every} \bullet \text{book}); \mathbf{V}_{\text{book}}; S \upharpoonright (S \upharpoonright \text{NP})$

(19)
$$\frac{\frac{\text{read}; \mathbf{read}; (\text{NP} \setminus S) / \text{NP} \quad [\varphi_1; x; \text{NP}]^1}{\text{read} \bullet \varphi_1; \mathbf{read}(x); \text{NP} \setminus S} \upharpoonright^E \quad \frac{\text{john}; \mathbf{j}; \text{NP}}{\text{john} \bullet \text{read} \bullet \varphi_1; \mathbf{read}(x)(\mathbf{j}); S} \upharpoonright^E}{\frac{\text{john} \bullet \text{read} \bullet \varphi_1; \mathbf{read}(x)(\mathbf{j}); S}{\lambda\varphi_1.\text{john} \bullet \text{read} \bullet \varphi_1; \lambda x.\mathbf{read}(x)(\mathbf{j}); S \upharpoonright \text{NP}} \upharpoonright^1 \quad \frac{\vdots}{\lambda\sigma_1[\sigma_1(\text{every} \bullet \text{book})]; \mathbf{V}_{\text{book}}; S \upharpoonright (S \upharpoonright \text{NP})} \upharpoonright^E}$$

$$\frac{\lambda\sigma_1[\sigma_1(\text{every} \bullet \text{book})](\lambda\varphi_1.\text{john} \bullet \text{read} \bullet \varphi_1); \mathbf{V}_{\text{book}}(\lambda x.\mathbf{read}(x)(\mathbf{j})); S}{\dots \lambda\varphi_1[\text{john} \bullet \text{read} \bullet \varphi_1](\text{every} \bullet \text{book}); \mathbf{V}_{\text{book}}(\lambda x.\mathbf{read}(x)(\mathbf{j})); S} \upharpoonright^E$$

$$\frac{\dots \lambda\varphi_1[\text{john} \bullet \text{read} \bullet \varphi_1](\text{every} \bullet \text{book}); \mathbf{V}_{\text{book}}(\lambda x.\mathbf{read}(x)(\mathbf{j})); S}{\text{john} \bullet \text{read} \bullet \text{every} \bullet \text{book}; \mathbf{V}_{\text{book}}(\lambda x.\mathbf{read}(x)(\mathbf{j})); S}$$

The analysis of quantifier scope in (19) involves a technique called ‘hypothetical reasoning’. Specifically, we first hypothetically assume the existence of some NP in the object position (with prosody φ_1), and once a complete sentence is formed, withdraw that hypothesis (the penultimate step marked $\upharpoonright^1$). The derived linguistic expression, of type **st** $\rightarrow$ **st** prosodically (i.e., a function from strings to strings) and $S \upharpoonright \text{NP}$ Syntactically, is essentially a sentence that is missing an NP in the object position. This ‘gapped sentence’ is then fed to the quantifier, which lowers its string component (i.e., the string $\text{every} \bullet \text{book}$) to the ‘gap’ position corresponding to the variable φ_1 by function application (the dotted lines show the reduction steps for the prosodic term for the readers’ convenience, but these steps are inessential for the purpose of the proof). In the semantics, the GQ operator ($\mathbf{V}_{\text{book}} = \lambda P.\forall x[\mathbf{book}(x) \rightarrow P(x)]$) simply takes the property denoted by the ‘open proposition’ of type $S \upharpoonright \text{NP}$ (semantically, $e \rightarrow t$) as an argument.

2 and Appendix A) for a more detailed exposition of the formal calculus.

The key assumption here is that in this QR/quantifying-in analysis of GQs, the scopal operator is assigned a functional term not only in the semantics but also in the prosodic representation as well. This approach straightforwardly extends to modals with the lexical entry for the modal auxiliary *can* in (20), formally implementing the analytic idea graphically represented in (16). Note that (20) resembles the entry for the quantifier in (18) (here, VP_f is an abbreviation for $NP \setminus S_f$; similarly for VP_b).

$$(20) \quad \lambda\sigma.\sigma(\text{can}); \lambda\mathcal{F}.\Diamond\mathcal{F}(\mathbf{id}_{et}); S_f \upharpoonright (S_f \upharpoonright (VP_f/VP_b))$$

$$(21) \quad \frac{\lambda\sigma.\sigma(\text{can}); \lambda\mathcal{F}.\Diamond\mathcal{F}(\mathbf{id}_{et}); S_f \upharpoonright (S_f \upharpoonright (VP_f/VP_b)) \quad \textcircled{D} \rightarrow \frac{\frac{\frac{\text{john}; \mathbf{j}; NP \quad [\varphi_1; f; VP_f/VP_b]^1 \quad \text{swim}; \mathbf{swim}; VP_b}{\varphi_1 \bullet \text{swim}; f(\mathbf{swim}); VP_f} /E}{\text{john} \bullet \varphi_1 \bullet \text{swim}; f(\mathbf{swim})(\mathbf{j}); S_f} \setminus E}{\frac{\lambda\varphi_1.\text{john} \bullet \varphi_1 \bullet \text{swim}; \lambda f.f(\mathbf{swim})(\mathbf{j}); S_f \upharpoonright (VP_f/VP_b)}{\lambda\sigma[\sigma(\text{can})](\lambda\varphi_1.\text{john} \bullet \varphi_1 \bullet \text{swim}); \Diamond\mathbf{swim}(\mathbf{j}); S_f} \upharpoonright I^1} \upharpoonright E$$

$$\frac{\lambda\sigma[\sigma(\text{can})](\lambda\varphi_1.\text{john} \bullet \varphi_1 \bullet \text{swim}); \Diamond\mathbf{swim}(\mathbf{j}); S_f}{\lambda\varphi_1[\text{john} \bullet \varphi_1 \bullet \text{swim}](\text{can}); \Diamond\mathbf{swim}(\mathbf{j}); S_f} \dots$$

$$\frac{\lambda\varphi_1[\text{john} \bullet \varphi_1 \bullet \text{swim}](\text{can}); \Diamond\mathbf{swim}(\mathbf{j}); S_f}{\text{john} \bullet \text{can} \bullet \text{swim}; \Diamond\mathbf{swim}(\mathbf{j}); S_f} \dots$$

Just as with GQs, a hypothetical VP/VP is posited in the ‘surface’ position of the modal auxiliary. This hypothesis is withdrawn to yield an expression of type $S \upharpoonright (VP/VP)$ at the clausal level (step $\textcircled{D}$), which is then given as an argument to the higher-order modal of type $S \upharpoonright (S \upharpoonright (VP/VP))$ to complete the derivation (again, the dotted lines are just for convenience). Semantically, the main contribution of the modal is the propositional operator $\Diamond$, and the variable f corresponding to the type VP/VP preverbal gap is filled in by $\mathbf{id}_{et}$, that is, the identity function $\lambda P_{et}.P$ for expressions of type $e \rightarrow t$.

Since some of the proofs below are somewhat complex, we introduce a simpler (but notationally equivalent) format of proofs here, building on the theoretical result known as Curry-Howard Isomorphism (Howard 1969). The key idea is that there is a lambda calculus

for the syntactic logic (which is related to, but is distinct from the lambda calculi used for specifying the semantic and prosodic information of each linguistic sign) for writing proofs. This syntactic lambda calculus distinguishes three types of function application ($\text{app}_/$, $\text{app}_\backslash$, and $\text{app}_\uparrow$), and three types of lambda abstraction ($\lambda_/_$, $\lambda_\backslash$, and $\lambda_\uparrow$). Then, an Elimination step in the natural deduction proof ($/E$, $\backslash E$ or $\uparrow E$), that is, a step at which a function combines with its argument, corresponds to function application in the lambda calculus, and an Introduction step ($/I$, $\backslash I$ or $\uparrow I$), where a hypothesis posited at an earlier step in the proof is withdrawn, corresponds to lambda abstraction in the syntactic lambda calculus.

As an illustration, we show in (22b) the lambda term encoding of the proof in (21), using the abbreviatory notations for tripartite linguistic signs in (22a).

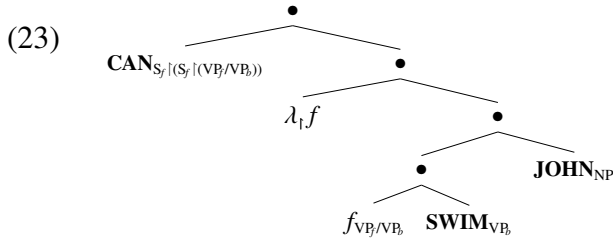
$$\begin{aligned}
(22) \quad & \text{a. } \mathbf{CAN}_{S_f \uparrow (S_f \uparrow (VP_f / VP_b))} = \lambda \sigma. \sigma(\text{can}); \lambda \mathcal{F}. \Diamond \mathcal{F}(\mathbf{id}_{et}); S_f \uparrow (S_f \uparrow (VP_f / VP_b)) \\
& \mathbf{SWIM}_{VP_b} = \text{swim}; \mathbf{swim}; VP_b \\
& \mathbf{JOHN}_{NP} = \text{john}; \mathbf{j}; NP \\
& \text{b. } \mathbf{CAN}_{S_f \uparrow (S_f \uparrow (VP_f / VP_b))}(\lambda_\uparrow f. f_{VP_f / VP_b}(\mathbf{SWIM}_{VP_b})(\mathbf{JOHN}_{NP})) \\
& = \lambda \sigma[\sigma(\text{can})](\lambda \varphi_1. \text{john} \bullet \varphi_1 \bullet \text{swim}); \lambda \mathcal{F}[\Diamond \mathcal{F}(\mathbf{id}_{et})](\lambda f. f(\mathbf{swim})(\mathbf{j})); S_f \\
& = \text{john} \bullet \text{can} \bullet \text{swim}; \Diamond \mathbf{swim}(\mathbf{j}); S_f
\end{aligned}$$

In (22b), the variable f_{VP_f / VP_b} takes two arguments $\mathbf{SWIM}_{VP_b}$ and $\mathbf{JOHN}_{NP}$. These two function applications correspond to the two Elimination steps in the proof tree combining the hypothetical VP_f / VP_b with its two arguments in (21). In the natural deduction proof, this

is followed by an Introduction step withdrawing the hypothesis, and this corresponds to the lambda abstraction of f in the proof term notation in (22b). Note that a variable in the proof term corresponds to a hypothesis in the natural deduction notation. For notational ease, we have chosen the same variable letter f for the variable in the syntactic proof term in (22b) and in the semantic term calculus in (21), but it is important to keep in mind that these are formally distinct variables in different calculi. Finally, the higher-order modal *can* takes the derived expression of type $S_f \upharpoonright (VP_f/VP_b)$ as an argument via $\upharpoonright E$, which again corresponds to function application in the proof term (the variety of application rule is omitted in (22b), since this information is unambiguously recoverable from the syntactic type of the function). The reduction of the term in the prosodic and semantic components in (22b) should make it clear that the derived type S_f sign is identical to the one in (21).

Note that there is also a one-to-one correspondence between the types of the lambda terms and ‘formulas’ (i.e., syntactic types/categories) derived at each step in the natural deduction proof. For example, the subformula $\lambda_{\upharpoonright} f. f_{VP_f/VP_b}(\mathbf{SWIM}_{VP_b})(\mathbf{JOHN}_{NP})$ in (22b) corresponding to the subtree up to the $\upharpoonright I$ step in (21) is of type $S_f \upharpoonright (VP_f/VP_b)$ since $f_{VP_f/VP_b}(\mathbf{SWIM}_{VP_b})(\mathbf{JOHN}_{NP})$ is of type S_f . In this sense, the term in (22b) represents the complete proof (21) in the term calculus for the type-logic of Hybrid TLG.

One advantage of the proof term notation in (22b) is that it makes the structure of the proof transparent to read, hiding much of the inessential visual clutter in (21). We can make this point even clearer by drawing the lambda term in (22b) in the form of a tree, as in (23).

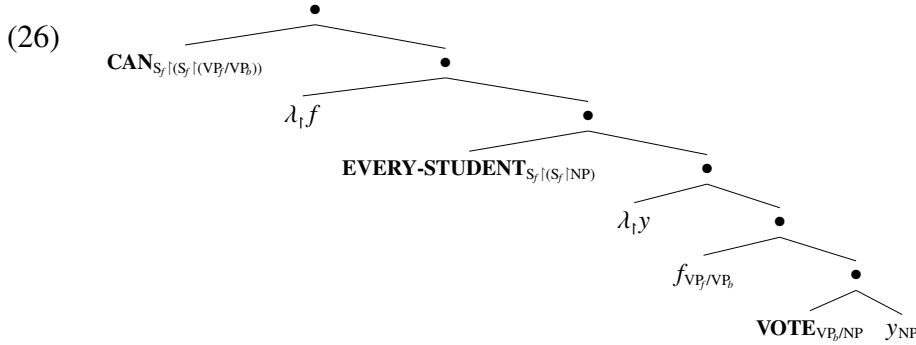


This is especially useful for the purpose of seeing how one and the same string gets associated with different scopal relations among operators. We illustrate this with the scopal relation between modals and subject position quantifiers in what follows.

The proof for the wide-scope reading for the modal is shown in (25), with the sign for *every student* in (24) (the natural deduction proof is given in (73) in Appendix B).

(24) $\mathbf{EVERY-STUDENT}_{S_f \upharpoonright (S_f \upharpoonright NP)} = \lambda \sigma_1. \sigma_1(\text{every} \bullet \text{student}); \mathbf{V}_{\text{student}}; S_f \upharpoonright (S_f \upharpoonright NP)$

(25) $\mathbf{CAN}_{S_f \upharpoonright (S_f \upharpoonright (V_B/V_B))}(\lambda_1 f. \mathbf{EVERY-STUDENT}_{S_f \upharpoonright (S_f \upharpoonright NP)}(\lambda_1 y. f_{V_B/V_B}(\mathbf{VOTE}_{V_B})(y_{NP})))$
 $= \text{every} \bullet \text{student} \bullet \text{can} \bullet \text{vote}; \Diamond \mathbf{V}_{\text{student}}(\lambda y. \mathbf{vote}(y)); S_f$

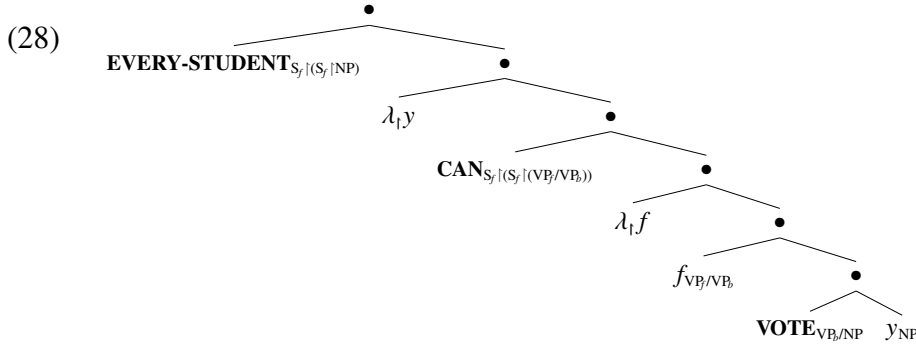


As should be clear from the structure of the proof term, the quantifier is introduced in the proof before the auxiliary is. Since the relative scope between these operators in the semantic translation reflects this inside-outside relation in the proof structure, we obtain the reading in which the modal auxiliary outscopes the quantifier, as the reduced term on the second line of (25) makes clear.

The quantifier-wide scope reading is obtained in the following proof, yielding the same string but with a different semantic interpretation:

$$(27) \quad \mathbf{EVERY-STUDENT}_{S_f \upharpoonright (S_f \upharpoonright \text{NP})}(\lambda_1 y. \mathbf{CAN}_{S_f \upharpoonright (S_f \upharpoonright (\text{VP}_f/\text{VP}_b))}(\lambda_1 f. f_{\text{VP}_f/\text{VP}_b}(\mathbf{VOTE}_{\text{VP}_b})(y_{\text{NP}})))$$

$$= \text{every} \bullet \text{student} \bullet \text{can} \bullet \text{vote}; \mathbf{V}_{\text{student}}(\lambda y. \Diamond \mathbf{vote}(y)); S_f$$



Here again, the proof structure transparently corresponds to the relative scope relations between the operators. Note that, unlike lexicalist approaches in which lexical entries need to be duplicated, in our approach, the scope ambiguity is predicted by just positing the single higher-order lexical entry for the modal auxiliary in (20).

The clause-level scope analysis illustrated above immediately predicts the anomalous scope pattern of modals in Gapping sentences. The proof for the modal wide scope reading for (4a) is shown in (29) (here, $\mathbf{GC}(\mathcal{T}) = \mathcal{T} \upharpoonright \mathcal{T} \upharpoonright \mathcal{T}$).

$$(29) \quad \mathbf{SHOULDN'T}_{S_f \upharpoonright (S_f \upharpoonright (\text{VP}_f/\text{VP}_b))}(\lambda_1 h.$$

$$\mathbf{AND}_{\mathbf{GC}(S_f \upharpoonright (\text{VP}_f/\text{NP}))}(\lambda_1 f. f_{\text{VP}_f/\text{NP}}(\mathbf{STEAK}_{\text{NP}})(\mathbf{JOHN}_{\text{NP}}))$$

$$(\lambda_1 g. g_{\text{VP}_f/\text{NP}}(\mathbf{PIZZA}_{\text{NP}})(\mathbf{MARY}_{\text{NP}}))$$

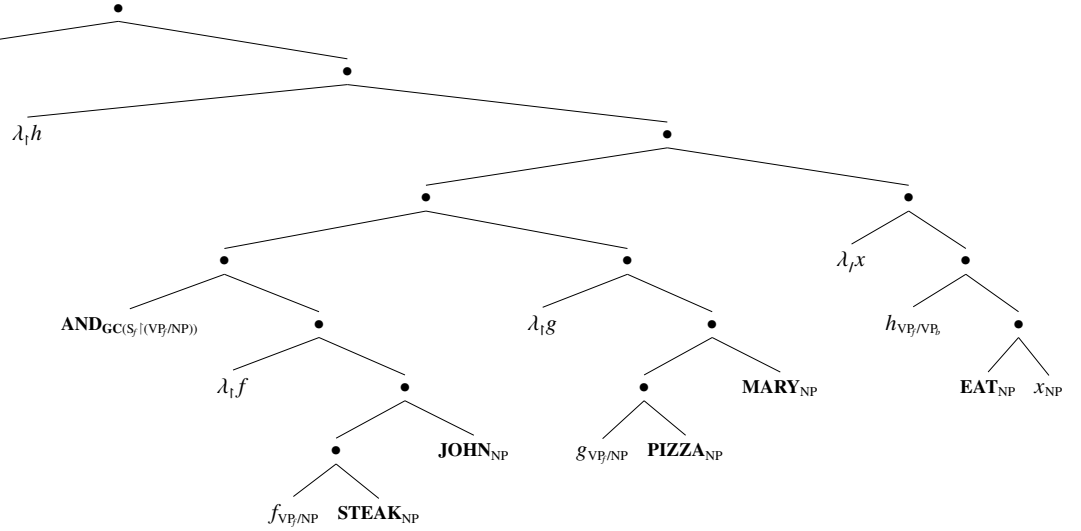
$$(\lambda_1 x. h_{\text{VP}_f/\text{VP}_b}(\mathbf{EAT}_{\text{VP}_b/\text{NP}}(x_{\text{NP}}))))$$

= john • shouldn't • eat • steak • and • mary • pizza;

$\Box \neg [\text{eat}(\mathbf{s})(\mathbf{j}) \wedge \text{eat}(\mathbf{p})(\mathbf{m})]; S_f$

(30)

SHOULDN'T $T_{S_f \upharpoonright (VP/VP)}$



As in (31), the prosodic term for $\text{AND}_{GC(S_f \upharpoonright (VP/NP))}$ specifies that the gap of type VP/NP of the second conjunct is filled in by an empty string and only the gap of the first conjunct gets filled in by an overt material.

(31) $\text{AND}_{GC(S_f \upharpoonright (VP/NP))} =$

$\lambda \sigma_2 \lambda \sigma_1 \lambda \varphi. \sigma_1(\varphi) \bullet \text{and} \bullet \sigma_2(\epsilon);$

$\lambda \mathcal{F}_2 \lambda \mathcal{F}_1. \mathcal{F}_1 \sqcap \mathcal{F}_2; (S_f \upharpoonright (VP_f/NP)) \upharpoonright (S_f \upharpoonright (VP_f/NP)) \upharpoonright (S_f \upharpoonright (VP_f/NP))$

In (29), the TV-type gap is first filled in by a transitive verb that itself has a VP/VP gap (the variable h in the proof term). Binding this gap yields an expression of type $S \upharpoonright (VP/VP)$, which can be given as an argument to the modal auxiliary to complete the proof of S_f (see Kubota and Levine (2020, 65–66) for a more detailed exposition). The structure of the proof term makes it clear that the modal outscopes conjunction in semantic interpretation.

It should be clear from the above that the principled account of the scopal interactions with other operators that modal auxiliaries exhibit is driven by what one might call a ‘pseudo-covert-movement effect’ that falls out, with no additional machinery, as an immediate consequence of hypothetical reasoning with the vertical slash in Hybrid TLG. Since hypothetical reasoning is a very general and flexible mode of logical inference, unless appropriately constrained the present analysis has the danger of overgenerating readings for modals (and related expressions such as VP negation) in which they scope out of their local clauses:

(32) #John thinks that Ann should buy the car.

‘John should think that Ann buys the car’

One way of preventing this type of overgeneration is to employ the type of meta-constraint on the forms of syntactic derivations that Kubota and Levine (2025) employ for the purpose of analyzing extraction pathway marking.³ Since we do not deal with examples involving multiple clausal embedding in the rest of the present paper, we will not comment on this issue further and just assume that something along the lines of footnote 3 prevents the kind of overgeneration exemplified by (32).

³What needs to be done is to prevent a VP_f/VP_b type gap from being unbound at a clause boundary. This can be done by imposing the following constraint on the complementizer:

- (i) **THAT** = that; $\lambda f.f; S'/S$
 where for any α , **THAT**(α) is well-defined only if $\forall x \in fV(\alpha). \mathbf{SynTyp}(x) \neq VP_f/VP_b$

This blocks a structure like the following in which an embedded gap of type VP_f/VP_b remains unbound and gets passed on to the higher clause:

- (ii) **SHOULD** _{$S_f \downarrow (S_f \downarrow (VP_f/VP_b))$} ($[\lambda f.f. \mathbf{THINK}_{VP_f/S'}$
 $\mathbf{THAT}_{S'/S_f}(f_{VP_f/VP_b}(\mathbf{BUY-THE-CAR}_{VP_b})(\mathbf{ANN}_{NP}))(\mathbf{JOHN}_{NP})]$ _{$S_f \downarrow (VP_f/VP_b)$})

Returning to the payoffs of the higher-order analysis of modal auxiliaries, one interesting consequence that falls out from the lexical entry in (20) and the logical calculus of Hybrid TLG is that a ‘surface-oriented’ VP/VP entry for a modal auxiliary identical to (7) in the previous section is obtained as a theorem. We here show the proof in both the natural deduction format in (33) and the proof term format in (34).

$$\begin{aligned}
 (33) \quad & \frac{\lambda\sigma.\sigma(\text{can't}); \quad \lambda\mathcal{F}.\neg\Diamond\mathcal{F}(\mathbf{id}_{et}); \quad S_f \upharpoonright (S_f \upharpoonright (VP_f/VP_b))}{\frac{\frac{[\varphi_1; x; NP]^1 \quad \frac{[\varphi_2; g; VP_f/VP_b]^2 \quad [\varphi_3; f; VP_b]^3}{\varphi_2 \bullet \varphi_3; g(f); VP_f} \backslash E}{\varphi_1 \bullet \varphi_2 \bullet \varphi_3; g(f)(x); S_f} \backslash E}{\lambda\varphi_2.\varphi_1 \bullet \varphi_2 \bullet \varphi_3; \lambda g.g(f)(x); S_f \upharpoonright (VP_f/VP_b)} \upharpoonright I^2} \upharpoonright E \\
 & \frac{\frac{\frac{\varphi_1 \bullet \text{can't} \bullet \varphi_3; \neg\Diamond f(x); S_f}{\text{can't} \bullet \varphi_3; \lambda x.\neg\Diamond f(x); VP_f} \upharpoonright I^1}{\text{can't}; \lambda f \lambda x.\neg\Diamond f(x); VP_f/VP_b} / I^3} \\
 (34) \quad & \lambda_l f \lambda_{\lambda} x. \mathbf{CAN'T}_{S_f \upharpoonright (S_f \upharpoonright (VP_f/VP_b))} (\lambda_l g \cdot g_{VP_f/VP_b} (f_{VP_b}) (x_{NP})) \\
 & = \text{can't}; \lambda f \lambda x.\neg\Diamond f(x); VP_f/VP_b
 \end{aligned}$$

The proof makes extensive use of hypothetical reasoning (which corresponds to lambda abstraction in the proof term notation), and may look somewhat abstract initially. But the key idea here is very simple and straightforward. Three hypotheses g_{VP_f/VP_b} , f_{VP_b} and x_{NP} are first combined to yield an S_f . The hypothesis g_{VP_f/VP_b} is then withdrawn to derive $S_f \upharpoonright (VP_f/VP_b)$, which is a suitable argument for the modal auxiliary **CAN'T**. This is followed by two steps of lambda abstraction, removing the prosodic variables φ_1 and φ_3 in that order. The natural deduction proof makes it clear that these steps are licitly licensed by the Introduction rules $\backslash I$ and $/I$, since the respective prosodic variables occupy the right and left edges of the prosodic representations of the premises.⁴

⁴The proof term notation hides this aspect, but here too, the applicability condition on lambda abstraction by λ_l and λ_{λ} is determined according to how the (beta-reduced) prosodic component of the body of the lambda term looks, essentially referring to the same information as the natural deduction notation.

Stepping back from the details, we can make sense of the whole proof in (33)/(34) as follows. The function of the original higher-order lexical entry for the auxiliary in (20) is to ‘fill in’ the missing VP/VP. The above proof essentially shows that the effect of this higher-order modal can be ‘compiled out’ to a simpler, surface-oriented lexical entry that directly works as the ‘missing’ VP/VP.

It should then be not too surprising that the more complex, ‘subject GQ’ lexical entry (cf. (12)) also falls out as a theorem. This time, we show the proof in the proof term notation only, in order to highlight the key similarities and differences with (34) in a compact notation (see (74) in Appendix B for a natural deduction proof).

$$(35) \quad \lambda_l f \lambda_{\downarrow} \mathcal{F} . \mathbf{CAN}' \mathbf{T}_{S_f \downarrow (S_f \downarrow (VP_f / VP_b))} (\lambda_{\downarrow} g . \mathcal{F}_{S_f / VP_f} (g_{VP_f / VP_b} (f_{VP_b})))$$

$$= \text{can}' t; \lambda f \lambda \mathcal{F} . \neg \Diamond \mathcal{F}(f); ((S_f / VP_f) \setminus S_f) / VP_b$$

Note that the only difference from (34) is that in (35), instead of hypothesizing an NP in the subject position, we hypothesize a Lambek-type GQ expression of type $S/(NP \setminus S)$ (which is of type $(e \rightarrow t) \rightarrow t$ semantically).

One way of viewing the relationship between our approach and previous lexicalist approaches is that the more abstract approach to syntax we have outlined above offers a kind of meta-theory for the type of ‘psycholinguistically plausible’, surface-oriented grammar that advocates of lexicalist approaches propose.⁵ For example, the two Lambek-type lexical entries in (34) and (35) (i.e. those involving only / and $\setminus$, and not $\downarrow$) for the modal corresponding

⁵A word of caution is in order here regarding the relationship between competence grammar and a theory of performance, since matters are quite complex and subtle in reality; Wasow (2021)—a major proponent of monostratal theories—explicitly notes that the arguments adduced for such theories based on ‘psycholinguistic plausibility’ have, for the most part, been entirely conceptual.

to different scope interpretations for the subject position quantifiers are both derived from a single lexical entry in (20) in our approach. We are not claiming that such abstract computations are carried out in the brain each time sentences with modals are processed by a human parser. Rather, the point here is that the meta-theory allows us to capture generalizations about the theoretical constructs posited in the surface-oriented theory in a way that is (in our view) ultimately illuminating (see also Jäger (2005) in this connection). One might still contend that such a meta-theory is superfluous (see Chaves and Putnam (2020, section 4.4) for a particularly clearly expressed version of such a view, directed against the current transformational approach). We offer below our own response to this reasonable skepticism. We argue that our approach offers a conceptually simple account of the NI(C)E properties and the notorious *do* support facts as well, with interesting crosslinguistic and diachronic implications.

3.2 A higher-order operator analysis of NIE properties

Among the four NICE properties, contraction exhibits a certain degree of morphological idiosyncrasy, thus justifying a lexical treatment. For this reason, we set it aside in the rest of this paper. For the other three phenomena, our analysis builds on the key idea that auxiliary verbs are syntactically operators that fill in the preverbal gap position of type VP/VP. Specifically, we analyze NIE operators as higher-order operators that manipulate the VP/VP-type gap before it gets filled in by the lexical auxiliary.

3.2.1 Basic actions of the NIE operators

The word-order change in subject-auxiliary inversion can simply be handled by positing the following higher-order operator that maps a $S \downarrow (VP/VP)$ to another $S \downarrow (VP/VP)$ which differs only in the prosodic specification. The fact that inversion has taken place is recorded in the syntactic feature *inv*, a standard technique for distinguishing inverted from non-inverted clauses in lexicalist approaches (here, ϵ designates the empty string).⁶

$$(36) \quad \lambda\sigma\lambda\varphi.\varphi \bullet \sigma(\epsilon); \lambda\mathcal{F}.\mathcal{F}; (S_{inv} \downarrow (VP_f/VP_b)) \downarrow (S_f \downarrow (VP_f/VP_b))$$

In the prosodic component, the first argument σ is a function of type **st**→**st**. This function is converted to a function of the same type (**st**→**st**), but one which concatenates its argument at the beginning of the clause in which the original ‘gap’ position is closed off by the empty string.

After the inversion operator applies to $S \downarrow (VP/VP)$, the result is passed on to the higher-order auxiliary. The latter fills in the auxiliary string in the gap position—which has been moved to the clause-initial position by the inversion operator—to complete the derivation (here, we have slightly generalized the lexical entry for the auxiliary, replacing S_f with S_α , where $\alpha \in \{fin, inv\}$).

$$(37) \quad \begin{array}{c} \vdots \\ \lambda\sigma.\sigma(\text{should}); \quad \lambda\sigma\lambda\varphi.\varphi \bullet \sigma(\epsilon); \quad \lambda\varphi.\text{john} \bullet \varphi \bullet \text{come}; \\ \lambda\mathcal{F}.\square\mathcal{F}(\text{id}_{et}); \quad \lambda\mathcal{F}.\mathcal{F}; \quad \lambda f.f(\text{come})(j); \\ S_\alpha \downarrow (S_\alpha \downarrow (VP_f/VP_b)) \quad (S_{inv} \downarrow (VP_f/VP_b)) \downarrow (S_f \downarrow (VP_f/VP_b)) \quad S_f \downarrow (VP_f/VP_b) \\ \hline \lambda\varphi.\varphi \bullet \text{john} \bullet \text{come}; \lambda f.f(\text{come})(j); S_{inv} \downarrow (VP_f/VP_b) \\ \hline \text{should} \bullet \text{john} \bullet \text{come}; \square\text{come}(j); S_{inv} \end{array}$$

⁶The semantics of the inversion operator in (36) is the identity function. Since inversion is not limited to interrogative sentences, we assume that a separate operator (of the sort assumed in Kubota and Levine (2021)) is responsible for introducing the semantics of polar questions.

An important fact about auxiliary inversion is that it is a clause-bound phenomenon.

(38) *Will anybody who ___ be vaccinated after arrival may visit Japan?

We utilize the clause-level indexing mechanism from the previous section to capture this fact explicitly. Specifically, (36) is actually an abbreviation of (39), which makes it explicit that the ‘auxiliary gap’ that the inversion operator targets is a local one.

(39) $\lambda\sigma\lambda\varphi.\varphi \bullet \sigma(\epsilon); \lambda\mathcal{F}.\mathcal{F}; (S_{inv}^n \downarrow (VP_f^n/VP_b^n)) \downarrow (S_f^n \downarrow (VP_f^n/VP_b^n))$

We make the same assumption about the ellipsis and negation operators below, but continue to suppress the clause-level indexing for the sake of readability.

Moving on to ellipsis, descriptively, VP ellipsis is a phenomenon in which an auxiliary stands in for a full VP. Our analysis in (40) is modelled after the lexicalist analysis in (10) from the previous section. The key idea that VP ellipsis involves valence reduction (that is, a VP/VP stands in for a VP) is retained here, but the analysis is adjusted in such a way that the form of the operator is more parallel to the inversion operator introduced above.

(40) $\lambda\sigma\lambda\varphi.\sigma(\varphi); \lambda\mathcal{G}\lambda f.\mathcal{G}(f(P)); (S_f \downarrow (VP_f/VP_b)) \downarrow (S_b \downarrow VP_b)$

Note that the prosody of the operator is just the identity function for type **st**→**st** prosodic expressions, and in the semantics a free variable P is thrown in as the meaning of the missing VP.

A sample derivation is given in (41). Just as in the case of inversion, the ellipsis operator applies first and the resultant $S \downarrow (VP/VP)$ expression is passed on to the higher-order auxiliary.

$$(41) \quad \frac{\begin{array}{c} \lambda\sigma.\sigma(\text{should}); \\ \lambda\mathcal{F}.\Box\mathcal{F}(\mathbf{id}_{et}); \\ S_\alpha \upharpoonright (S_\alpha \upharpoonright (VP_f/VP_b)) \end{array} \quad \frac{\begin{array}{c} \lambda\sigma\lambda\varphi.\sigma(\varphi); \\ \lambda\mathcal{G}\lambda f.\mathcal{G}(f(P)); \\ (S_f \upharpoonright (VP_f/VP_b)) \upharpoonright (S_b \upharpoonright VP_b) \end{array} \quad \begin{array}{c} \vdots \\ \lambda\varphi.\text{john} \bullet \varphi; \\ \lambda\mathcal{Q}.\mathcal{Q}(\mathbf{j}); S_b \upharpoonright VP_b \end{array}}{\lambda\varphi.\text{john} \bullet \varphi; \lambda f.f(P)(\mathbf{j}); S_f \upharpoonright (VP_f/VP_b)} \\ \hline \text{john} \bullet \text{should}; \Box P(\mathbf{j}); S_f$$

Finally, negation can be analyzed as a higher-order operator similar to the modal auxiliaries as in (42).

$$(42) \quad \lambda\sigma.\sigma(\text{not}); \lambda\mathcal{F}.\neg\mathcal{F}(\mathbf{id}_{et}); S_\alpha \upharpoonright (S_\alpha \upharpoonright (VP_b/VP_b))$$

This may not look similar to the inversion and ellipsis operators introduced above, but an alternative sign that applies to a auxiliary-gapped sentence and inserts the negation morpheme right after the gap can be obtained as a theorem as in (43).

$$(43) \quad \lambda_{\upharpoonright}\mathcal{F}\lambda_{\upharpoonright}f.\mathcal{F}_{S_f \upharpoonright (VP_f/VP_b)}(\lambda_{\upharpoonright}P\lambda_{\upharpoonright}x.\mathbf{NOT}_{S_\alpha \upharpoonright (S_\alpha \upharpoonright (VP_b/VP_b))}(\lambda_{\upharpoonright}g.f_{VP_f/VP_b}(g_{VP_b/VP_b}(P_{VP}))(x_{NP}))) \\ = \lambda\sigma\lambda\varphi_1.\sigma(\varphi_1 \bullet \text{not}); \lambda\mathcal{F}\lambda f.\mathcal{F}(\lambda P\lambda x.\neg f(P)(x)); (S_f \upharpoonright (VP_f/VP_b)) \upharpoonright (S_f \upharpoonright (VP_f/VP_b))$$

This more complex sign in (43) plays a crucial role in our analysis of the interactions of the NIE operators in the next section.

(44) illustrates a derivation in which the derived negation operator in (43) takes a gapped sentence of type $S \upharpoonright (VP/VP)$ and maps it to a ‘negated’ $S \upharpoonright (VP/VP)$ that can be given as an argument to the higher-order auxiliary in a way parallel to the working of inversion and ellipsis operators above.

$$(44) \quad \frac{\begin{array}{c} \vdots \\ \lambda\sigma.\sigma(\text{should}); \\ \lambda\mathcal{F}.\Box\mathcal{F}(\mathbf{id}_{et}); \\ S_\alpha \upharpoonright (S_\alpha \upharpoonright (VP_f/VP_b)) \end{array} \quad \frac{\begin{array}{c} \vdots \\ \lambda\sigma\lambda\varphi_1.\sigma(\varphi_1 \bullet \text{not}); \\ \lambda\mathcal{F}\lambda f.\mathcal{F}(\lambda P\lambda x.\neg f(P)(x)); \\ (S_f \upharpoonright (VP_f/VP_b)) \upharpoonright (S_f \upharpoonright (VP_f/VP_b)) \end{array} \quad \begin{array}{c} \vdots \\ \lambda\varphi.\text{john} \bullet \varphi \bullet \text{come}; \\ \lambda f.f(\mathbf{come})(\mathbf{j}); \\ S_f \upharpoonright (VP_f/VP_b) \end{array}}{\lambda\varphi.\text{John} \bullet \varphi \bullet \text{not} \bullet \text{come}; \lambda f.\neg f(\mathbf{come})(\mathbf{j}); S_f \upharpoonright (VP_f/VP_b)} \\ \hline \text{john} \bullet \text{should} \bullet \text{not} \bullet \text{come}; \Box \neg \mathbf{come}(\mathbf{j}); S_f$$

3.2.2 NIE Interactions

To facilitate the ensuing discussion, we introduce some abbreviatory notation: **INV**, **ELL** and **NEG** for the three operators introduced above and **LEX** for some auxiliary lexical entry (*should* is chosen here just for an illustration; $\Rightarrow$ is ‘generalized negation’ such that $\Rightarrow_{et \rightarrow et} = \lambda \mathcal{F} \lambda P \lambda x. \neg \mathcal{F}(P)(x)$).⁷

- (45) a. **INV** = $\lambda \sigma \lambda \varphi. \varphi \bullet \sigma(\epsilon); \lambda \mathcal{F}. \mathcal{F}; (S_{inv} \upharpoonright (VP_f/VP_b)) \upharpoonright (S_f \upharpoonright (VP_f/VP_b))$
- b. **ELL** = $\lambda \sigma \lambda \varphi. \sigma(\varphi); \lambda \mathcal{G} \lambda f. \mathcal{G}(f(P)); (S_f \upharpoonright (VP_f/VP_b)) \upharpoonright (S_b \upharpoonright VP_b)$
- c. **NEG** = $\lambda \sigma \lambda \varphi. \sigma(\varphi \bullet \text{not});$
- $\lambda \mathcal{F} \lambda g. \mathcal{F}(\Rightarrow_{et \rightarrow et} g); (S_f \upharpoonright (VP_f/VP_b)) \upharpoonright (S_f \upharpoonright (VP_f/VP_b))$
- d. **LEX** = $\lambda \sigma. \sigma(\text{should}); \lambda \mathcal{F}. \Box \mathcal{F}(\mathbf{id}_{et}); S_\alpha \upharpoonright (S_\alpha \upharpoonright (VP_f/VP_b))$

Given these abbreviatory notations, we can derive the inverted, complement-elided and negated versions of the auxiliary lexical signs as follows, via function composition of **LEX** and the three operators (see (75) in Appendix B for a complete natural deduction proof for (46a); other theorems in (46) can be proved similarly):

- (46) a. **LEX** $\circ$ **INV** = $\lambda \mathcal{F}. \mathbf{LEX}(\mathbf{INV}(\mathcal{F}))$
- $= \lambda \sigma. \text{should} \bullet \sigma(\epsilon); \lambda \mathcal{F}. \Box \mathcal{F}(\mathbf{id}_{et}); S_{inv} \upharpoonright (S_f \upharpoonright (VP_f/VP_b))$
- b. **LEX** $\circ$ **ELL** = $\lambda \mathcal{F}. \mathbf{LEX}(\mathbf{ELL}(\mathcal{F}))$

⁷To accommodate the parallel analysis of VP ellipsis and pseudogapping in Kubota and Levine (2017), the ellipsis operator in (45) should be generalized along the lines of (i), where the missing verbal element can be of a polymorphic type. Since we do not deal with pseudogapping, we gloss over this refinement in the rest of this paper.

(i) $\lambda \sigma \lambda \varphi. \sigma(\varphi); \lambda \mathcal{G} \lambda f. \mathcal{G}(f(P)); (S_f \upharpoonright ((VP_f/\$)/(VP_b/\$))) \upharpoonright (S_b \upharpoonright (VP_b/\$))$

$$= \lambda \sigma. \sigma(\text{should}); \lambda \mathcal{G}. \Box \mathcal{G}(P); S_f \upharpoonright (S_b \upharpoonright VP_b)$$

$$\text{c. } \mathbf{LEX} \circ \mathbf{NEG} = \lambda_1 \mathcal{F}. \mathbf{LEX}(\mathbf{NEG}(\mathcal{F}))$$

$$= \lambda \sigma. \sigma(\text{should} \bullet \text{not}); \lambda \mathcal{F}. \Box \mathcal{F}(\Rightarrow_{et \rightarrow et}(\mathbf{id}_{et})); S_f \upharpoonright (S_f \upharpoonright (VP_f / VP_b))$$

One interesting consequence is that a ‘surface-oriented’ version of the inverted auxiliary sign is obtained as a theorem (a natural deduction proof is in (76) in Appendix B):

$$(47) \quad \lambda_I x \lambda_J P. (\mathbf{LEX} \circ \mathbf{INV})_{S_{inv} \upharpoonright (S_f \upharpoonright (VP_f / VP_b))} (\lambda_1 f. f_{VP_f / VP_b} (P_{VP_b})(x_{NP}))$$

$$= \text{should}; \lambda x \lambda P. \Box P(x); S_{inv} / VP_b / NP$$

Here again, our approach can be thought of as a proper generalization (in a formally explicit system) of the familiar analysis in the lexicalist tradition.

The interactions of the NIE phenomena can be captured by the interactions of the three operators. Since the proofs are all straightforward, we omit them (except for (48e)) but just list the relevant composed operators that are involved in each of the NIE interactions in (48e–h).

(48)	a. John should come.	LEX
	b. Should John come?	LEX $\circ$ INV
	c. John should $\emptyset$.	LEX $\circ$ ELL
	d. John should not come.	LEX $\circ$ NEG
	e. Should John not come?	LEX $\circ$ INV $\circ$ NEG
	f. John should not $\emptyset$.	LEX $\circ$ NEG $\circ$ ELL

g. Should John?

LEX $\circ$ **INV** $\circ$ **ELL**

h. Should John not $\emptyset$?

LEX $\circ$ **INV** $\circ$ **NEG** $\circ$ **ELL**

Just as an illustration, (49) shows the proof for **LEX** $\circ$ **INV** $\circ$ **NEG** used in (48e), and (50) a derivation for (48e) using (49).

$$\begin{aligned}
 (49) \quad & \mathbf{LEX} \circ \mathbf{INV} \circ \mathbf{NEG} = \lambda_{\uparrow} \mathcal{F} . \mathbf{LEX}(\mathbf{INV}(\mathbf{NEG}(\mathcal{F}))) \\
 &= \lambda \sigma [\lambda \sigma [\sigma(\text{should})] (\lambda \sigma \lambda \varphi [\varphi \bullet \sigma(\epsilon)] (\lambda \sigma \lambda \varphi [\sigma(\varphi \bullet \text{not})] (\sigma)))]; \\
 & \quad \lambda \mathcal{F} [\lambda \mathcal{F} [\square \mathcal{F}(\mathbf{id}_{et})] (\lambda \mathcal{F} [\mathcal{F}] (\lambda \mathcal{F} \lambda g [\mathcal{F}(\neg_{et \rightarrow et} g)] (\mathcal{F})))]]; S_{inv} \upharpoonright (S_f \upharpoonright (VP_f / VP_b)) \\
 &= \lambda \sigma . \text{should} \bullet \sigma(\text{not}); \lambda \mathcal{F} . \square \mathcal{F}(\neg_{et \rightarrow et} \mathbf{id}_{et}); S_{inv} \upharpoonright (S_f \upharpoonright (VP_f / VP_b))
 \end{aligned}$$

$$\begin{aligned}
 (50) \quad & \text{a. } \mathbf{JOHN-COME}_{S_f \upharpoonright (VP_f / VP_b)} \\
 &= \lambda_{\uparrow} f . f_{VP_f / VP_b}(\mathbf{COME}_{VP_b})(\mathbf{JOHN}_{NP}) \\
 &= \lambda \varphi . \text{john} \bullet \varphi \bullet \text{come}; \lambda \mathcal{F} . \mathcal{F}(\mathbf{come})(\mathbf{j}); S_f \upharpoonright (VP_f / VP_b) \\
 & \text{b. } (\mathbf{LEX} \circ \mathbf{INV} \circ \mathbf{NEG})(\mathbf{JOHN-COME}_{S_f \upharpoonright (VP_f / VP_b)}) \\
 &= \lambda \sigma [\text{should} \bullet \sigma(\text{not})] (\lambda \varphi . \text{john} \bullet \varphi \bullet \text{come}); \\
 & \quad \lambda \mathcal{F} [\square \mathcal{F}(\neg_{et \rightarrow et} \mathbf{id}_{et})] (\lambda \mathcal{F} . \mathcal{F}(\mathbf{come})(\mathbf{j})); S_{inv} \\
 &= \text{should} \bullet \text{john} \bullet \text{not} \bullet \text{come}; \square \neg \mathbf{come}(\mathbf{j}); S_{inv}
 \end{aligned}$$

Note the obvious connection to lexicalist and constructional approaches (such as Kim and Sag (2002) and Sag et al. (2020)). In such approaches, the NIE interactions are treated by letting the relevant lexical rules or constructional schemata feed into one another. Here, the idea is the same, except that in our case the interactions of the relevant operators are

governed by the same logic of syntactic combinatorics (in which function composition is a theorem) that governs other aspects of syntax.

3.3 The distribution of the unstressed *do*

The curious property of unstressed *do* is that its distribution is limited to environments in which an auxiliary *must* appear. Positing the following lexical entry for *dō/dīd* that is essentially identical to modals immediately leads to overgeneration of the declarative case (here, **P** is the past tense operator).

$$(51) \quad \mathbb{L}_{\text{EX}} = \lambda\sigma.\sigma(\text{dīd}); \lambda\mathcal{F}.\mathbf{P}\mathcal{F}(\mathbf{id}_{et}); S_\alpha \upharpoonright (S_\alpha \upharpoonright (\text{VP}_f/\text{VP}_b))$$

In our setup, there is a conceptually simple solution for this problem that cannot be easily translated to a PSG (or transformational) setup. Specifically, we take ‘*do* support’ to be mediated by an operator that takes NIE operators as arguments to produce the same effect as the ‘phantom’ (i.e. non-existent) lexical entry for *dō/dīd* in (51) above. In other words, we posit **Do** which satisfies the following property:

$$(52) \quad \text{For } f \in \{\mathbf{NEG}/\mathbf{ELL}/\mathbf{INV}\}, \mathbf{Do}(f) \equiv \mathbb{L}_{\text{EX}} \circ f$$

Given that we know what $\mathbb{L}_{\text{EX}}$ is (see (51)), defining **Do** is straightforward:

$$\begin{aligned} (53) \quad \mathbf{Do} &= \lambda f. \mathbb{L}_{\text{EX}} \circ f \\ &= \lambda f \lambda g. \mathbb{L}_{\text{EX}}(f(g)) \\ &= \lambda \rho \lambda \sigma. [\lambda \sigma_0. \sigma_0(\text{dīd})](\rho(\sigma)); \lambda \mathcal{G} \lambda h. [\lambda \mathcal{F}. \mathbf{P}\mathcal{F}(\mathbf{id}_{et})](\mathcal{G}(h)); (S_\alpha \upharpoonright \mathbf{X}) \upharpoonright (S_\alpha \upharpoonright (\text{VP}_f/\text{VP}_b) \upharpoonright \mathbf{X}) \\ &= \lambda \rho \lambda \sigma. \rho(\sigma)(\text{dīd}); \lambda \mathcal{G} \lambda h. \mathbf{P}\mathcal{G}(h)(\mathbf{id}_{et}); (S_\alpha \upharpoonright \mathbf{X}) \upharpoonright (S_\alpha \upharpoonright (\text{VP}_f/\text{VP}_b) \upharpoonright \mathbf{X}) \end{aligned}$$

where $X \in \{S_f \upharpoonright (VP_f/VP_b), S_b \upharpoonright VP_b\}$

Given the definition in (53), it should be straightforward to see that we get the right result in the NIE sentences. For example, in the inversion case, we have:

$$\begin{aligned}
 (54) \quad \mathbf{Do}(\mathbf{Inv}) &= \lambda f. [\mathbb{L}_{\mathbf{EX}} \circ f](\mathbf{Inv}) \\
 &= \mathbb{L}_{\mathbf{EX}} \circ \mathbf{Inv} \\
 &= \lambda \sigma. \text{d}\ddot{\text{id}} \bullet \sigma(\epsilon); \lambda \mathcal{F}. \mathbf{P} \mathcal{F}(\mathbf{id}_{et}); S_{inv} \upharpoonright (S_f \upharpoonright (VP_f/VP_b))
 \end{aligned}$$

The key point here is that **Do** closes off the VP/VP gap by directly *applying to* the NIE operators (instead of function composing with them). This means that it can’t work alone, from which it immediately follows that (2a) is not licensed. In this respect, the present proposal is reminiscent of the idea of ‘last resort’ that constitutes the underlying intuition of various formulations of *do* support in derivational approaches since Chomsky (1957), without the need to make any appeal to an inherently transderivational notion. In the present logic-based setup, there is a conceptually simple and mathematically precise way of formalizing this operation by means of a higher-order operator.

In our approach, the logic underlying syntactic combinatorics and that governing ‘lexical/morphological’ operations (such as the NIE operators and ‘*do* support’) are one and the same. One might then observe that, at least at the broader conceptual level, our approach has more in common with approaches such as Distributed Morphology (Halle and Marantz 1993) than with lexicalist approaches that categorial grammar has traditionally been associated with (note also in this connection that recent ‘constructional’ avatars of HPSG represented by Sign-based Construction Grammar similarly obliterates the traditional distinction

between the lexicon and syntax). While a detailed comparison with the DM perspective is beyond the scope of the present paper (see, e.g., Embick and Noyer (2001) for an analysis of *do* support in DM), we would like to highlight some potential differences. Most importantly, we do not assume syntactic representations in the form of trees at the morphological level (or, for that matter, at the syntactic level either). Relatedly, the functional operations we assume for doing morphology do not seem to have any direct analog in DM or its variants. Thus, to the extent that taking the inference-based view we advocate constitutes an advantage, our approach and DM are not mere notational variants.

3.4 Lexical idiosyncrasies

One question that arises at this point is how to treat lexical idiosyncrasies. This question is important in view of the major dispute between transformational and nontransformational approaches in the current literature. For example, Sag et al. (2020), a representative work in the latter tradition, explicitly identifies such idiosyncrasies as major challenges to transformational approaches (such as Pollock 1989) that attempt to reduce the properties of the auxiliary system to abstract general principles. While it is beyond the scope of the present paper to provide a comprehensive treatment of lexical idiosyncrasies, we sketch here possible solutions for some well-known cases.

We start with the case of inverted *shall*, which has a deontic force lacking in its uninverted counterpart (cf., e.g., Gazdar et al. 1985), and the use of *aren't* with a first person singular subject that is licensed only in the uninverted environment.

(55) a. Shall I inform John of our decision?

b. I shall inform John of our decision.

(56) a. Aren't I clever?!

b. *I aren't clever.

Such cases can be handled in a way strictly parallel to their treatment in previous lexicalist approaches. For example, an additional entry for *shall* in (57), whose type is $S_{inv} \uparrow (S_f \uparrow (VP_f/VP_b))$, can be used to license (55a).

(57) $\lambda\sigma.\text{shall} \bullet \sigma(\epsilon); \lambda\mathcal{F}.\Box.\mathcal{F}(\mathbf{id}_{et}); S_{inv} \uparrow (S_f \uparrow (VP_f/VP_b))$

Note that this is identical—except for semantics—to **SHALL** $\circ$ **INV**, where **SHALL** is the standard lexical entry for non-deontic *shall* of type $S_\alpha \uparrow (S_\alpha \uparrow (VP_f/VP_b))$. That is, (57) is essentially an inversion-specific lexical entry which already incorporates the effect of **INV**. A similar treatment is possible for *aren't* in (56a).

The form *better* in (58) is another case in point.

(58) a. You better (not)(say that).

b. *Better you (not)(say that)?

Negation, ellipsis and their composition are all possible, but inversion is ill-formed. The distributional restriction on *better* is essentially a mirror image of the restriction on deontic *shall* and the first person singular *aren't*. The same strategy of restricting the distribution via feature specification works here. Specifically, the contrast in (58) can be captured by making **BETTER** specifically target S_f -rooted clauses, as in (59).

$$(59) \quad \mathbf{BETTER} = \lambda\sigma.\sigma(\text{better}); \lambda\mathcal{F}.\square_{\mathbf{better}}\mathcal{F}(\mathbf{id}_{et}); S_f \upharpoonright (S_f \upharpoonright (VP_f/VP_b))$$

From this, it immediately follows that $\mathbf{BETTER} \circ \mathbf{NEG}$ and $\mathbf{BETTER} \circ \mathbf{ELL}$ are well-formed, but $*\mathbf{BETTER} \circ \mathbf{INV}$ isn't, the latter being blocked via feature conflict ($\mathbf{BETTER}$ is looking for $S_f \upharpoonright X$, but $\mathbf{INV}$ returns $S_{inv} \upharpoonright X$).⁸

Thus, lexical idiosyncrasies can be captured by restricting the combinations of operations via syntactic features or by positing expressions that are similar in form to the result of compositionally analyzable items (such as (57) for $\mathbf{SHALL} \circ \mathbf{INV}$), both familiar techniques in the tradition of nontransformational syntax. In this way, it encompasses within a single framework both generality of the sort characteristic of transformational approaches and mechanisms for dealing with specific idiosyncrasies that are the hallmark of certain contemporary versions of nontransformational syntax.

4 Crosslinguistic and diachronic implications

We now take a somewhat broader perspective and explore briefly the crosslinguistic and diachronic implications of our proposal. Our approach potentially offers a new angle on some of the classical and well-known problems in these domains too, here again synthesizing analytic ideas that have previously been taken to belong to different traditions.

⁸The limited distribution of *better* makes sense once we take into account its likely historical origin, namely, the contracted form *'d better* for *had better*. Note crucially that (58b) cannot have any such origin—**'d better you say that?*. It then make sense that *better* hasn't acquired a full-fledged status of an auxiliary, but still comes with the idiosyncratic restriction of S_f for S_α in the general schema for English auxiliaries in (20).

4.1 English vs. French

One of the key motivations for Pollock's (1989) configurational analysis was to account for the typological difference between English and French via the parametric variation in the presence vs. absence of V-to-T raising, exemplified by data such as (60)–(62).

- (60) a. *John likes not Mary.
b. Jean (**n'**) aime **pas** Marie.
- (61) a. *Likes he Mary?
b. Aime-t-il Marie?
- (62) a. *John kisses often Mary.
b. Jean embrasse **souvent** Marie.
c. John often kisses Mary.
d. *Jean **souvent** embrasse Marie.

In a nutshell, the account goes as follows. Both languages have underlying syntactic structures that look like the following (reproduced from section 2.2):

- (63) [CP C [TP [T' T [NegP [Neg *not*] [AgrP Agr [VP (Adv) V]]]]]]

Then, on the assumption that the subject NP stays in the VP-internal position in polar questions, the contrast between English and French is straightforward, by making the following single assumption: in French, a lexical verb raises from V to T in surface syntactic structure, but in English, it doesn't (see section 2.2 for more details).

A type-logical reinterpretation of the head movement analysis by Pollock is available in our approach by taking finite verbs in French to be higher-order functions that simultaneously bind the base-form verb gap position and the auxiliary gap position. The relevant lexical entry can be formulated as in (64).

$$(64) \quad \lambda\sigma.\sigma(\epsilon)(\text{aime}); \lambda f.f(\mathbf{love})(\mathbf{id}_{et}); S_f \downarrow (S_f \downarrow (VP_f/VP_b)) \downarrow (VP_b/NP))$$

The higher-order entry for the verb *aime* in (64) takes a sentence containing two gaps of type VP_b/NP and VP_f/VP_b . A transitive verb meaning and an identity function meaning are supplied in the corresponding λ -bound semantic variables, but prosodically, the overt finite form of the verb is embedded in the auxiliary gap position, with the original transitive verb gap position filled in by an empty string. This ‘flip’ of the two gap positions in the prosody has the effect of ‘verb raising’, as we show in (66) below.

The lexical entries for the negation and inversion operators are the same as in English (except that negation *n’ ... pas* in French is a circumfix), and we can simply assume that a VP adverb such as *souvent* are of type VP_b/VP_b .

$$(65) \quad \begin{aligned} \text{a. } \mathbf{INV}_{fr} &= \lambda\sigma\lambda\varphi.\varphi \bullet \sigma(\epsilon); \lambda\mathcal{F}.\mathcal{F}; (S_{inv} \downarrow (VP_f/VP_b)) \downarrow (S_f \downarrow (VP_f/VP_b)) \\ \text{b. } \mathbf{NEG}_{fr} &= \lambda\sigma\lambda\varphi.\sigma(n' \bullet \varphi \bullet \text{pas}); \\ &\quad \lambda\mathcal{F}\lambda g.\mathcal{F}(\exists_{et \rightarrow et} g); (S_f \downarrow (VP_f/VP_b)) \downarrow (S_f \downarrow (VP_f/VP_b)) \\ \text{c. } \text{souvent; often;} &VP_b/VP_b \end{aligned}$$

This fragment licenses the correct word-order patterns exhibited by (60)–(62). We illustrate this with the negation example (60) and the adverb example (62). The derivation for (60) goes as in (66).

$$(66) \quad \frac{\lambda\sigma\lambda\varphi.\sigma(n' \bullet \varphi \bullet \text{pas}); \quad \lambda\mathcal{F}\lambda g.\mathcal{F}(\Rightarrow_{et \rightarrow et} g); \quad (S_f \upharpoonright (\text{VP}_f/\text{VP}_b)) \upharpoonright (\text{S}_f \upharpoonright (\text{VP}_f/\text{VP}_b)) \quad \frac{\frac{\vdots \quad \vdots}{\text{jean} \bullet \varphi_0 \bullet \varphi_1 \bullet \text{marie}; \quad f(g(\mathbf{m}))(\mathbf{j}); S_f} \quad \frac{\lambda\varphi_0.\text{jean} \bullet \varphi_0 \bullet \varphi_1 \bullet \text{marie}; \quad \lambda f.f(g(\mathbf{m}))(\mathbf{j}); S_f \upharpoonright (\text{VP}_f/\text{VP}_b)}{\lambda\varphi_0.\text{jean} \bullet n' \bullet \varphi_0 \bullet \text{pas} \bullet \varphi_1 \bullet \text{marie}; \quad \lambda f.\Rightarrow_{et \rightarrow et} f(g(\mathbf{m}))(\mathbf{j}); S_f \upharpoonright (\text{VP}_f/\text{VP}_b)} \quad \frac{\lambda\varphi_1\lambda\varphi_0.\text{jean} \bullet n' \bullet \varphi_0 \bullet \text{pas} \bullet \varphi_1 \bullet \text{marie}; \quad \lambda g\lambda f.\Rightarrow_{et \rightarrow et} f(g(\mathbf{m}))(\mathbf{j}); S_f \upharpoonright (\text{VP}_f/\text{VP}_b) \upharpoonright (\text{VP}_b/\text{NP})}{\text{jean} \bullet n' \bullet \text{aime} \bullet \text{pas} \bullet \text{marie}; \neg\text{love}(\mathbf{m})(\mathbf{j}); S_f} \quad [\varphi_0; f; \text{VP}_f/\text{VP}_b]^0 \quad [\varphi_1; g; \text{VP}_b/\text{NP}]^1$$

The NEG_{fin} operator takes scope before the finite verb *aime* enters the derivation. This has the effect that the circumfix negative morpheme is attached to the auxiliary gap prosody φ_0 . The finite verb *aime* then takes this result and throws in the finite verb prosody in the auxiliary gap position while filling in an empty string in the original TV-type gap position. This yields the correct word order corresponding to (60b).

The adverb placement example (62) can be analyzed similarly. The derivation is given in (67).

$$(67) \quad \frac{\lambda \sigma. \sigma(\epsilon)(\text{embrasse}); \lambda f. f(\mathbf{kiss})(\mathbf{id}_{et}); S_f \upharpoonright (S_f \upharpoonright (VP_f / VP_b)) \upharpoonright (VP_b / NP)}{\text{jean} \bullet \text{embrasse} \bullet \text{souvent} \bullet \text{marie}; \mathbf{often}(\mathbf{kiss}(\mathbf{m}))(\mathbf{j}); S_f} \quad \frac{\frac{\text{jean}; \mathbf{j}; NP \quad \frac{\left[\varphi_0; f; VP_f / VP_b \right]^0}{\varphi_0 \bullet \text{souvent} \bullet \varphi_1 \bullet \text{marie}; f(\mathbf{often}(g(\mathbf{m}))); VP_f}}{\text{jean} \bullet \varphi_0 \bullet \text{souvent} \bullet \varphi_1 \bullet \text{marie}; f(\mathbf{often}(g(\mathbf{m}))) (\mathbf{j}); S_f}} \quad \frac{\text{souvent}; \mathbf{often}; VP_b / VP_b \quad \frac{\left[\varphi_1; g; VP_b / NP \right]^1}{\varphi_1 \bullet \text{marie}; g(\mathbf{m}); VP_b}}{\text{souvent} \bullet \varphi_1 \bullet \text{marie}; \mathbf{often}(g(\mathbf{m})); VP_b}}$$

Here, the auxiliary gap, the adverb and the base-form finite verb gap are linearly placed in this order given their syntactic types. The ‘verb raising’ higher-order entry of the finite verb again throws in its prosody in the auxiliary gap position rather than the original base form

An interesting consequence of this analysis is that the English-type ordinary transitive verb entry of type VP_f/NP can be derived as a theorem from the higher-order entry in (64). The derivation is shown in (68).

This should intuitively make sense given that, in the absence of negation, inversion or an adverb, the ‘movement’ of the finite verb from the base position to the auxiliary gap position is string vacuous, and the resulting SVO order is indistinguishable from the English word order.

We now turn to some diachronic aspects that are closely related to crosslinguistic variation of the sort discussed above. The discussion in this section builds heavily on work on historical syntax by other scholars—most importantly, by Anthony Warner (Warner 1993)—and our goal here is modest. We essentially endorse Warner’s view, but will argue that its key insights can be expressed even more transparently by taking a logic-based perspective of

the sort we have advocated above.

An important historical fact to keep in mind (see Warner (1993)) is that the development of *do* as an auxiliary took place right after the class of auxiliaries was established in English. The development of *do* as an auxiliary verb then took place essentially in two stages, as demonstrated with statistical analysis of corpus data in an influential study by Kroch (1989a). Kroch observes that from the beginning of the 16th century to the period of 1550–1575, the occurrence of *do* increases in all syntactic environments, including the declarative. But after the period of 1575–1600, the development in different contexts starts to diverge. *Do* in polar questions continues to increase but the use of *do* in declarative sentences starts to decline. In negative environments, there is an initial sharp decline, followed by a steady increase (but at a lower rate than in positive polar questions). Kroch attributes this two-stage change to the loss of V2 (or V-to-T raising; cf. Kroch (1989b)) in English within an account of parametric change in the Principles and Parameters (P&P) framework.

Warner offers a reinterpretation of Kroch's data that does not rely on the P&P assumptions (see also Hudson (1997)). According to Warner, the split in the development of *do* in different syntactic environments after 1575–1600 can be attributed to a reanalysis process. The key factor is the fact that the pragmatic function associated with *do* (which is essentially semantic focus on polarity contrast) is an inherent property of polar questions. Thus, with the steady increase of *do* in polar questions, the pragmatic function originally associated with *do* was reanalyzed as the constructional meaning of the polar interrogative itself, with *do* having only the function of contributing tense information.⁹ *Do* then

⁹As Warner notes, the initial steady increase of *do* in inverted structure is arguably related to the graduate

becomes essentially an ‘allomorph’ of the tense affix whose distribution is limited to the inverted interrogative environment. The use of periphrastic *do* in the declarative then declines gradually, via blocking by simple verb inflection for expressing tense. Note that this account is also consistent with the initial drop and the later recovery in negative environments; the reanalysis of *do* is triggered in the interrogative context, and then spreads to other syntactic contexts via analogy. Warner attributes the source of this reanalysis to child language learning. Due to the pragmatic function, *do* in the declarative environment was restricted to the literary style, but it was common in polar interrogatives in colloquial speech. Then, the child learning the language was most likely exposed to utterances containing the auxiliary use of *do* only in polar questions, from which s/he would infer that *do* is nothing more than a tense auxiliary.

One point that remains unclear in Warner’s account is the status of *do* as an ‘allomorph’ of the tense affix. Warner (1993, 250, n28) claims that a lexical specification involving the following implicational constraint (in HPSG) captures this generalization:

(69) SUBCAT $\langle [-\text{ELLIPSIS}, -\text{AUX}] \rangle$ & *do* $\supset$ semantic prominence

This says that an overt *do* carries the effect of ‘semantic prominence’ unless it appears in a NICE environment (SUBCAT $\langle [-\text{ELLIPSIS}, -\text{AUX}] \rangle$). However, this is just a restatement of the facts, and, perhaps more disturbingly, it is unclear how such a complex lexical restriction

decline of the V2 structure over the course of the Middle English period. Note that this is consistent with our reconceptualization of the Rich Agreement Hypothesis via an inflectional operator along the lines of (??) in the previous subsection. The loss of a rich inflectional paradigm on the verb means that there is less motivation for positing an abstract inflectional operator in the grammar, with finite verb forms simply being listed in the lexicon in the lower-type categories. As INFL declines in English, there is pressure to form interrogative sentences by an alternative strategy than fronting the inflected verb, giving rise to the increase of the focus-marking auxiliary *do* in polar questions.

arose out of a simple process of reanalysis of the sort Warner himself advocates.

Is there a simpler alternative for capturing the essential insight of Warner’s reanalysis-based account in a formal framework? We think there is. Specifically, within a type-logical setup like ours, the process can simply be understood as a reanalysis of LEX by **Do** from section 3.3. The situation can be schematically depicted as follows:

(70) **Adult grammar:**

- a. $\text{john } \varphi \text{ come} \xrightarrow{\text{LEX}} \text{john did come}$ (literary style)
- b. $\text{john } \varphi \text{ come} \xrightarrow{\text{LEX} \circ \text{INV}} \text{did john come}$ (colloquial)

(71) **Child grammar:**

- a. $\text{john } \varphi \text{ come} \xrightarrow{\quad} \text{john did come} \xrightarrow{\quad} \text{(literary style)}$
- b. $\text{john } \varphi \text{ come} \xrightarrow{??} \text{did john come}$ (colloquial)

In the first phase of the two-stage development of *do*, LEX used to exist in the grammar of English, so its distribution was not restricted to NICE environments, as in (70). But its usage was restricted to literary style in declaratives, due to the pragmatic function it was associated with. A child acquiring the language then gets exposed to occurrences of *do* only in polar questions as in (71).

This much is the same as in Warner’s account. But then, we essentially have a situation in which the child has to choose between two competing hypotheses, $\text{LEX} \circ \text{INV}$ vs. **Do**(INV), for solving the hidden operation ?? in (71b) (where the availability of INV is independently motivated by the distribution of modals). Here, the child lacks access to the crucial evi-

dence (i.e. (70a)) for assuming $\mathbb{L}_{\text{EX}}$; in other words, the available evidence gives him/her confidence only for the weaker hypothesis of positing the higher-order **Do**. The child then opts for this more conservative hypothesis. Note that, unlike in Warner’s original account (in which the status of the restriction (69) is unclear both conceptually and formally), the formal status of **Do** is perfectly explicit here: it is an operator that mimics the effect of an auxiliary just when the language doesn’t contain such an auxiliary but we nevertheless see its ‘effect’ in certain syntactic environments (specifically, in NIE environments). Moreover, our reconceptualization provides a clear motivation for why the reanalysis has happened: **Do** wins over $\mathbb{L}_{\text{EX}}$ in the acquisition context since it is the less risky hypothesis that is still consistent with all the data that the child encounters.¹⁰

Thus, the higher-order analysis of *do* support potentially illuminates the origin of this process as well. Of course, there is no way of directly proving or disproving the hypothesis we have entertained above. However, other things being equal, an analysis that provides a natural motivation for known facts about historical change is more preferable than one that doesn’t.

¹⁰Note that on this scenario, we have a straightforward explanation for how the hypothesized **Do** operator comes to display the same morphophonological signature as the lexical verb of action *do* (and also the emphatic auxiliary *do*). We hypothesize that in the transition phase from lexical auxiliary $\mathbb{L}_{\text{EX}}$ to **Do**, learners reinterpret each inflected instance of the former as an instance of the latter, so that, for example, *dīd* of type $S_f \downarrow (S_f \downarrow (VP_f/VP_b))$ with the ‘kernel’ string *dīd* is recast in type $(S_\alpha \downarrow X) \downarrow (S_\alpha \downarrow (VP_f/VP_b) \downarrow X)$ (similarly for *dīdn’t*, *dōes*, *dōesn’t*, etc.). The **Do** operator, then, without itself being subject to the inflectional rules of English morphology, nonetheless winds up preserving their output.

5 Conclusion

Our starting point in this paper was the ‘higher-order’ analysis of modal auxiliaries due to Kubota and Levine (2016, 2020), implemented with a lambda abstraction operation in the prosodic component in Hybrid TLG. An interesting property of this analysis is that it embodies a synthesis of the key analytic ideas associated with transformational and non-transformational approaches to syntax, that is, what have traditionally been regarded as two competing (and mutually incompatible) approaches in generative syntax research. The resultant analysis reveals the deeper connections between the two in a way that is difficult to appreciate otherwise. The payoff for this synthesis, we have argued, comes from the fact that it offers principled accounts of two key properties of English auxiliary verbs: their scopal interactions with other operators and the well-known NI(C)E properties and the *do* support facts.

As we see it, a major source of the long-standing contention between the transformational and nontransformational approaches is the tension between being (sufficiently) explicit and aiming to uncover higher-level abstract principles. What we have tried to emphasize throughout this paper is that this tension is illusory and that it is, in our view, largely an artefact of the ‘building-block’ view that has dominated syntactic research over the past 60 years. At least in the domain of the English auxiliary system, once we identify precisely the underlying ‘logic’ governing the complex distributional patterns on the surface, there is nothing that prevents one from being totally explicit and achieving generalizations at an abstract level at the same time. Moreover, taking the inference-based perspective has the

additional bonus of illuminating some of the crosslinguistic and diachronic aspects, as well as smoothly accommodating apparent challenges such as lexical idiosyncrasies. While all this may sound too good to be true, the empirical evidence that has accumulated in the rich research tradition in generative grammar seems to point precisely to such an architecture of grammar, along lines we have argued in this paper.

A Rules of Hybrid Type-Logical Grammar

(72)

Connective	Introduction	Elimination
/	$\frac{\begin{array}{c} \vdots \quad [\varphi; x; A]^n \quad \vdots \\ \vdots \quad \vdots \quad \vdots \\ \hline b \bullet \varphi; \mathcal{F}; B \\ b; \lambda x. \mathcal{F}; B/A \end{array}}{\Lambda^n}$	$\frac{a; \mathcal{F}; A/B \quad b; \mathcal{G}; B}{a \bullet b; \mathcal{F}(\mathcal{G}); A} /E$
\	$\frac{\begin{array}{c} \vdots \quad [\varphi; x; A]^n \quad \vdots \\ \vdots \quad \vdots \quad \vdots \\ \hline \varphi \bullet b; \mathcal{F}; B \\ b; \lambda x. \mathcal{F}; A \backslash B \end{array}}{\backslash^n}$	$\frac{b; \mathcal{G}; B \quad a; \mathcal{F}; B \backslash A}{b \bullet a; \mathcal{F}(\mathcal{G}); A} \backslash E$
↑	$\frac{\begin{array}{c} \vdots \quad [\varphi; x; A]^n \quad \vdots \\ \vdots \quad \vdots \quad \vdots \\ \hline b; \mathcal{F}; B \\ \lambda \varphi. b; \lambda x. \mathcal{F}; B \uparrow A \end{array}}{\uparrow^n}$	$\frac{a; \mathcal{F}; A \uparrow B \quad b; \mathcal{G}; B}{a(b); \mathcal{F}(\mathcal{G}); A} \uparrow E$

B Ancillary proofs

(73) modal/GQ interaction (= (25))

$$\begin{array}{c}
\frac{\text{vote;} \left[\begin{array}{c} \varphi_1; \\ f; \\ \text{VP}_f/\text{VP}_b \end{array} \right]^1}{\varphi_1 \bullet \text{vote}; f(\text{vote}); \text{VP}_f} \quad \left[\begin{array}{c} \varphi_2; \\ y; \\ \text{NP} \end{array} \right]^2 \quad \vdots \\
\frac{\varphi_2 \bullet \varphi_1 \bullet \text{vote}; f(\text{vote})(y); S_f}{\lambda \varphi_2. \varphi_2 \bullet \varphi_1 \bullet \text{vote}; \lambda y. f(\text{vote})(y); S_f \upharpoonright \text{NP}} \quad \frac{\lambda \sigma_1. \sigma_1(\text{every} \bullet \text{student}); \mathbf{V}_{\text{student}}; S_f \upharpoonright (S_f \upharpoonright \text{NP})}{\text{every} \bullet \text{student} \bullet \varphi_1 \bullet \text{vote}; \mathbf{V}_{\text{student}}(\lambda y. f(\text{vote})(y)); S_f} \\
\frac{\lambda \varphi_1. \text{every} \bullet \text{student} \bullet \varphi_1 \bullet \text{vote}; \lambda f. \mathbf{V}_{\text{student}}(\lambda y. f(\text{vote})(y)); S_f \upharpoonright (\text{VP}_f/\text{VP}_b)}{\text{every} \bullet \text{student} \bullet \text{can} \bullet \text{vote}; \Diamond \mathbf{V}_{\text{student}}(\lambda y. \text{vote}(y)); S_f} \quad \frac{\lambda \sigma_2. \sigma_2(\text{can}); \lambda \mathcal{F}. \Diamond \mathcal{F}(\mathbf{id}_{et}); S_f \upharpoonright (S_f \upharpoonright (\text{VP}_f/\text{VP}_b))}{\text{every} \bullet \text{student} \bullet \text{can} \bullet \text{vote}; \Diamond \mathbf{V}_{\text{student}}(\lambda y. \text{vote}(y)); S_f}
\end{array}$$

(74) subject GQ modal (= (35))

$$\begin{array}{c}
\lambda \sigma. \sigma(\text{can't}); \quad \frac{[\varphi_2; g; \text{VP}_f/\text{VP}_b]^2 \quad [\varphi_3; f; \text{VP}_b]^3}{\varphi_2 \bullet \varphi_3; g(f); \text{VP}_f} /E \\
\lambda \mathcal{F}. \neg \Diamond \mathcal{F}(\mathbf{id}_{et}); \quad \frac{[\varphi_1; \mathcal{F}; S_f/\text{VP}_f]^1 \quad \frac{\varphi_1 \bullet \varphi_2 \bullet \varphi_3; \mathcal{F}(g(f)); S_f}{\lambda \varphi_2. \varphi_1 \bullet \varphi_2 \bullet \varphi_3; \lambda g. \mathcal{F}(g(f)); S_f \upharpoonright (\text{VP}_f/\text{VP}_b)}}{\varphi_1 \bullet \text{can't} \bullet \varphi_3; \neg \Diamond \mathcal{F}(f); S_f} \wedge E \\
\frac{S_f \upharpoonright (S_f \upharpoonright (\text{VP}_f/\text{VP}_b)) \quad \lambda \varphi_2. \varphi_1 \bullet \varphi_2 \bullet \varphi_3; \lambda g. \mathcal{F}(g(f)); S_f \upharpoonright (\text{VP}_f/\text{VP}_b)}{\text{can't} \bullet \varphi_3; \lambda \mathcal{F}. \neg \Diamond \mathcal{F}(f); (S_f/\text{VP}_f) \setminus S_f} \wedge I^1 \\
\frac{\text{can't}; \lambda f \lambda \mathcal{F}. \neg \Diamond \mathcal{F}(f); ((S_f/\text{VP}_f) \setminus S_f)/\text{VP}_b}{\text{can't}; \lambda f \lambda \mathcal{F}. \neg \Diamond \mathcal{F}(f); ((S_f/\text{VP}_f) \setminus S_f)/\text{VP}_b} /I^3
\end{array}$$

(75) inverted modal (= (46a))

$$\begin{array}{c}
\lambda \sigma \lambda \varphi. \varphi \bullet \sigma(\epsilon); \quad \left[\begin{array}{c} \sigma_1; \\ \mathcal{F}; \\ S_f \upharpoonright (\text{VP}_f/\text{VP}_b) \end{array} \right]^1 \quad \lambda \sigma. \sigma(\text{should}); \lambda \mathcal{G}. \Box \mathcal{G}(\lambda R. R); S \upharpoonright (S \upharpoonright (\text{VP}_f/\text{VP}_b)) \\
\frac{(S_{inv} \upharpoonright (\text{VP}_f/\text{VP}_b)) \upharpoonright (S_f \upharpoonright (\text{VP}_f/\text{VP}_b)) \quad \lambda \varphi. \varphi \bullet \sigma_1(\epsilon); \mathcal{F}; S_{inv} \upharpoonright (\text{VP}_f/\text{VP}_b)}{\text{should} \bullet \sigma_1(\epsilon); \Box \mathcal{F}(\lambda R. R); S_{inv}} \\
\frac{\lambda \sigma_1. \text{should} \bullet \sigma_1(\epsilon); \lambda \mathcal{F}. \Box \mathcal{F}(\lambda R. R); S_{inv} \upharpoonright (S_f \upharpoonright (\text{VP}_f/\text{VP}_b))}{\lambda \sigma_1. \text{should} \bullet \sigma_1(\epsilon); \lambda \mathcal{F}. \Box \mathcal{F}(\lambda R. R); S_{inv} \upharpoonright (S_f \upharpoonright (\text{VP}_f/\text{VP}_b))} \wedge I^1
\end{array}$$

(76) ‘slanted’ inverted modal (= (47))

$$\begin{array}{c}
\text{LEX} \circ \text{INV} \\
\vdots \\
\lambda \sigma. \text{should} \bullet \sigma(\epsilon); \quad \frac{[\varphi_3; x; \text{NP}]^3 \quad \frac{[\varphi_1; f; \text{VP}_f/\text{VP}_b]^1 \quad [\varphi_2; P; \text{VP}_b]^2}{\varphi_1 \bullet \varphi_2; f(P); \text{VP}_f}}{\varphi_3 \bullet \varphi_1 \bullet \varphi_2; f(P)(x); S_f} \\
\lambda \mathcal{F}. \Box \mathcal{F}(\mathbf{id}_{et}); \quad \frac{S_{inv} \upharpoonright (S_f \upharpoonright (\text{VP}_f/\text{VP}_b)) \quad \lambda \varphi_1. \varphi_3 \bullet \varphi_1 \bullet \varphi_2; \lambda f. F(P)(x); S_f \upharpoonright (\text{VP}_f/\text{VP}_b)}{\text{should} \bullet \varphi_3 \bullet \varphi_2; \Box P(x); S_{inv}} \wedge I^1 \\
\frac{\text{should} \bullet \varphi_3; \lambda P. \Box P(x); S_{inv}/\text{VP}_b}{\text{should}; \lambda x \lambda P. \Box P(x); S_{inv}/\text{VP}_b/\text{NP}} /I^3
\end{array}$$

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